Synthetic Smartphone Data Usage Modeling Through User Clustering

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***Abstract*—Mobile network usage models (along with system models) are useful tools for estimating the performance of future and current mobile communication systems. However current mobile network usage models lack either the granularity or flexibility required for assessing advanced concepts (such as cognitive radio and dynamic spectrum access) at academic research levels. Therefore, in this work, we propose a network usage model that is both granular (modeling at the individual device level) and flexible (allowing arbitrary combinations of empirically derived user types and various time scales). We derive the user types through time-series clustering of network usage patterns of around 400 diverse US-based smartphone users. Additionally, the model accounts for the bursty nature and self-similarity of such usage and we show these characteristics are preserved over several time scales.**

I. Introduction

Early-stage development of new mobile communication systems require high-level assessment of candidate technologies prior to defining standards and protocols. In many cases this involves simulations of the new system or parts therein. Therefore, the realistic simulation of user usage also plays an important role. Although there exists a variety of commercial, open-source, and academic usage (traffic) simulators these are typically not derived from empirical measurements of individual users (user level) or very flexible. Moreover, especially for academic research purposes, models are required that produce realistic usage samples but do not require detailed modeling of the full protocol stack.

In this paper, we present an approach for simulating data usage of mobile networks based on characteristic activity patterns of different user types.[1] Such data can be obtained easily by analyzing the data usage of mobile users at session level, which motivates their usage for the establishment of demand simulation models. In particular, we focus on the creation of synthetic usage which displays the essential characteristics of data usage such as self-similarity, burstiness, and long-range dependence [1], but also accounts for the distinctiveness of individual user types. We design the model such that it produces usage samples adhering to these characteristics, and at the same time incorporates all relevant information that can be deduced from user based usage volume measurements.

The paper is organized as follows. After describing the related work and the used panel data (recording mobile data usage) in section II, we extract individual temporal usage patterns from the dataset (section III), and use them to parameterize a self-similar usage simulator, which is presented in section IV. In the same section, we also outline the setup of an exemplary scenario to verify the model's ability to reproduce the characteristics of the original dataset. The results are presented in section V, and section VI concludes.

II. Background

A. Related Work

Related work can be primarily classified as work with a similar data usage (traffic) modeling goal or work with similar clustering and/or modeling methods.

Ref. [2] models the temporal data usage at single or aggregated base stations (BSs) through Fourier analysis and distribution fittings of empirical data from a large city in China. Similarly, [3] models the highly heterogeneous spatial data usage across BSs based on empirical network data from a large 28000 km^2 area in China. Finally, [4] combines spatiotemporal modeling of data usage at BSs (through a non-separable co-variance function) again based on empirical network data from a large city in China. However, as mentioned, these studies focus on modeling at the BS level rather than the user level. Thus these studies are unable to identify and leverage diverse user types for more flexible modeling.

In terms of methods, several papers have applied similar clustering or distribution fitting methods. Ref. [5] applied a similar time-series clustering method to identify common network data usage pattern groups from mobile network data. Though, they apply their

[1]The term activity refers to the intensity of network usage, which can be measured by the amount of sent and received data usage over time, typically on hourly basis.

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method to M2M traffic rather than consumer traffic. Ref. [6] applied heavy tailed distribution fitting to empirical mobile usage data, though with a focus on the distribution of mobile device usage time over different mobile applications. Mobile data usage was not considered.

B. Dataset

The main dataset is mobile network usage data from a subset of a large user panel arranged by Verto Analytics[2] in the United States in February 2015 (one month observation period). Users were recruited online and were surveyed through an initial recruitment survey to determine the devices they own. Users were then instructed to install custom monitoring apps to all of their applicable devices (specifically their smartphone, tablet, and/or PC). All users were paid for participation. All provided user data was anonymized with no personally identifying information.

For our subset we include only active android smartphone users[3] and the associated network usage data from their smartphone. We define a user as active if their smartphone was used on at least 21 days of the month. This day threshold represents a trade-off between the sample size (number of analyzed users) and sample definition (ensuring analyzed users are truly active throughout the month). Figure 1 illustrates the normalized number of users considered active with different day thresholds including the selected threshold of 21 days. Overall the subset consists of 410 Android smartphone users.

For reference we provide a summary of user demographic data for our subset along with demographic data for US smartphone users in general in Table I. For space considerations, we refer to [7] for further information about the large user panel including a discussion about general panel study limitations.

TABLE I
DEMOGRAPHIC SUMMARIES FOR SUBSET USERS (ACTIVE ANDROID SMARTPHONE USERS) AND US SMARTPHONE USERS.

Demographic	Subset Users	US Users[a]
Mean Age (Years) [Std. Dev.]	37.52 [12.31]	41.30 [15.08]
Gender (% Male)	25.76	50.08
Education (% w/ SC[b] or Less)	62.41	53.04
Marital Status (% Married)	43.02	48.96
HH[c] Income (% <50K USD)	68.08	40.72
Mean HH[c] [Std. Dev.] Size	3.01 [1.54]	3.05 [1.61]
Mean Children HH[c][Std. Dev.]	1.01 [1.23]	0.74 [1.27]
Race (% White)	68.32	71.53

[a] US smartphone user demographic data is from Pew Research survey (June-July 2015, subpopulation with smartphone $n = 1327$) [8].
[b] Some College
[c] Household

[2]http://www.vertoanalytics.com/

[3]We include only android users because only the Android API provides reliable information about the devices current network connection type.

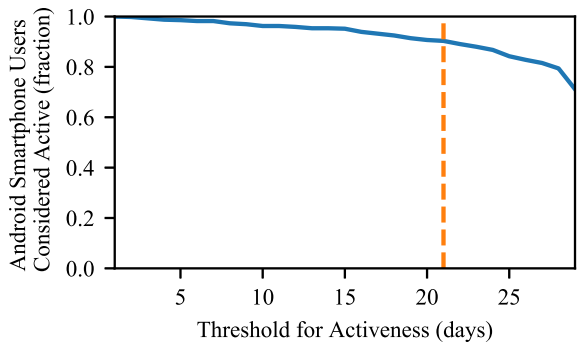

Fig. 1. Fraction of Android smartphone users considered active for different activeness thresholds in days (dashed line indicates chosen threshold of 21 days)

The raw smartphone usage data, for each user, consists of a collection of mobile network usage sessions. Each session represents the network usage by a specific application over a short interval of time and includes the following information: device id, current connected network type (Wifi, LTE, HSPA, etc.), application, start time, end time, sent data volume, and received data volume. Since we are only interested in modeling 3GPP network usage, we exclude all Wifi sessions. The monitoring app creates the sessions by monitoring the device's network interface and starting a new session (and potentially ending the previous) whenever the current connected network type changes, the user starts or ends an application process, the entire system is restarted, or the current session reaches a maximum length of 60 minutes.

III. PREPROCESSING

We first process the raw data usage sessions for each user into time series representing the mean daily per hour data volumes for combinations of *sent/received data* and *weekday/weekend*. Therefore, after this processing we have for each user four 24-element time series. We then normalize each time series by the mean daily data volume.

Then, for the weekday and weekend series separately, we perform a time-series clustering process similar to [5, Section IV]. We describe, as an example, the complete process for the weekday series (received and sent). We summarize the process for the weekend series afterwards.

First, for both the received data and sent data normalized time series, we calculate wavelet coefficients through discrete wavelet transform (with the Daubechies-1 wavelet and basis level of 3). Then we cluster users via agglomerative hierarchical clustering of these wavelet coefficients. We use Euclidean distance and ward linkages for the hierarchical clustering and select the optimal number of clusters via examination of Davies-Bouldin indexes and the elbow method. The

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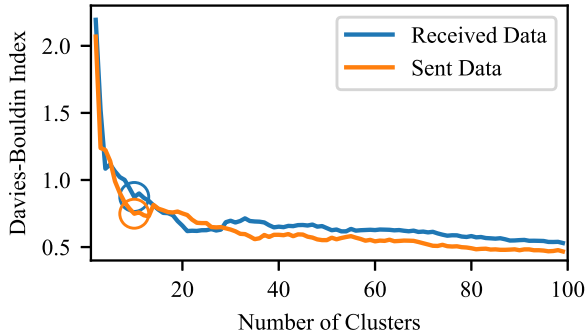

Fig. 2. Davies-Bouldin indexes for different numbers of clusters for *weekday* data usage (circles indicate the chosen number of clusters)

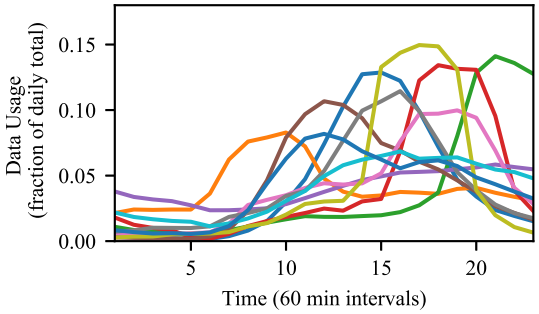

Fig. 3. Normalized user type activity patterns (NAPs) for *weekday received* data usage (5 hour moving average and excluding the std. dev. bars for visualization)

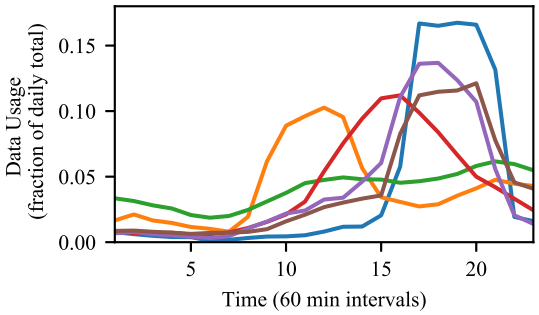

Fig. 4. Normalized user type activity patterns (NAPs) for *weekend received* data usage (5 hour moving average and excluding the std. dev. bars for visualization)

optimal number of clusters for received data is 10 and for sent data is 10. The Davies-Bouldin indexes for different numbers of clusters is illustrated in Figure 2. Then, users are classified into sent and received cluster combinations (of which there are 10×10 or 100 total). These combinations are hereafter referred to as user types. We discard 89 user types (retaining 11 types) with less than 10 users since further characterization of these infrequent types is unlikely to be reliable. These discarded infrequent types represent about 24% of users. The remaining user types have a total of 310 users.

For each remaining user type, we then calculate the centroids of the normalized sent and received series (with each centroid incorporating mean and standard deviation). Therefore, each user type is represented by two centroids that characterizes the most typical normalized daily received and sent data usage behavior of the users in the type. We hereafter refer to these normalized centroids as the normalized activity patterns of the user type (NAPs). We also later denote these NAPs as Ψ.

Since our NAPs are normalized, we still need to additionally model the actual daily data usage distribution for each user type. Through examination the distributions appear to be heavy tailed. Therefore we fit several well known heavy tail distributions (log-normal, exponential, stretched-exponential, power law, and truncated power law) to the data through maximum likelihood estimations [9]. We then select the best fit via Akaike weights. We find that the Weibull distribution[4] is the best fit for a plurality of user types and that the Weibull distribution is not significantly inferior ($p > 0.05$, Vuong likelihood ratio tests) to the best fit distributions for the remaining types. Therefore we use the Weibull distribution for all types. The best fit parameters for each user type are detailed in Table IV.

Similarly for weekend series, the optimal number of clusters for received data is 4 and for sent data is 8. Also we discard 26 infrequent user types (retaining 6 types) that represent about 13% of users. The remaining user types have a total of 357 users. Finally, we again use the Weibull distribution for all user types for similar reasons as in the weekday series. We note that due to the smaller number of clusters for weekend series versus weekday there are fewer weekend NAPs than weekday NAPs. Intuitively, this observation makes sense because of the larger diversity of daily working patterns (i.e. different work shifts) during weekdays compared to weekends.

For illustration, we show the NAPs for weekday and weekend received data usage in Figures 3 and 4. The NAPs for sent data usage are similar to the received activity patterns.

[4]For reference, the Weibull distribution PDF is $\lambda\beta(x\lambda)^{\beta-1}e^{-(x\lambda)^{\beta}}$ with fitted parameters λ and β.

IV. Synthetic Demand Model

Next we use the characterizations of user types to provide a generative model of data usage patterns. The model should be able to imitate behavioural patterns, in other words reproduce the empirical NAPs and distribution of daily received/sent volumes. Furthermore to imitate burstiness over various time scales, we additionally require the simulated data usage patterns to be self-similar.

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Self-similarity is achieved by modeling an $M/G/\infty$ queue as suggested by [10]. This approach considers Poisson-distributed burst arrivals and heavy-tailed burst sizes, which can be well approximated by the Pareto-distribution [11]. The approach can be used in combination with our empirical results as the NAPs can be interpreted as daily usage intensity patterns which can be related to the arrival rate λ of a non-homogeneous Poisson process. Therefore an empirical NAP modulates the burst arrival process. Limited by the time step size t_c of the empirical NAP, λ is determined by arbitrary interval sizes in the range of $0 < \Delta t \leq t_c$.

A. Activity Pattern Modeling

Synthetic usage will be generated for a number of K users at equally spaced time instances t_n, where $n \in \{0, 1, 2, ...\}$ and constant interval size $\Delta t = t_n - t_{n-1}$. Each user $k \in \{1, ..., K\}$ is member of a particular user type $c_k \in \{1, ..., C\}$, which determines its activity pattern $\Psi_k(t_n)$ (corresponding with its empirical NAP) and the parameters β_k and λ_k of the related daily received and sent usage volumes distribution (refer to Table IV). The simulation generates Weibull-distributed daily usage volumes separately for each user according to its user type membership. It also differentiates between weekdays and weekends. These volumes are used to generate the base sequences $v_{k,r}(t_n)$ and $v_{k,s}(t_n)$, respectively. Hereafter, the notation $v_{k,r/s}(t_n)$ refers to both, received and sent usage. Note, that within a day, these sequences remain constant. The same applies to $\Psi_{k,r/s}(t_n)$, which remain constant for all time steps within an hour. By combining the volume sequences with the activity patterns, we obtain an estimate for the intensity parameter λ of a non-homogeneous Poisson arrival process. We define the intensity as the mean number of packet bursts per time increment.

$$\lambda_{k,r/s}(t_n) = \Psi_{k,r/s}(t_n) \cdot v_{k,r/s}(t_n) \cdot \frac{\Delta t}{\mu_{r/s,burst}} \quad (1)$$

We need to allow for different mean burst sizes ($\mu_{r/s,burst}$) for received and sent usage [12]. The number of bursts per interval can thus be simulated individually for each user on the basis of two non-homogeneous Poisson processes $b_{k,r/s}(t_n) \sim Po(\lambda_{k,r/s}(t_n))$. Note, that the intensity of the processes vary on an hourly basis. However, even if $\Delta t << t_c$ (here: 1 hour), self-similarity is still preserved. Unless $\lambda_{k,r/s}(t_n)$ approaches 1, the model can be expected to produce a sufficiently large number of bursts per interval. This can be achieved by choosing smaller burst sizes and/or larger intervals. We obtained plausible values in the range of 1 min $\leq \Delta t \leq$ 60 min.

B. Burst Size Modeling

Usage bursts can be interpreted as sequences of data packets originating from a transmitting unit, i.e. either a mobile device (sent data) or a base station (received data). Although the distribution of burst lengths depends on the type of data and protocols used, aggregated data (i.e. over multiple flows of the same device or multiple devices) can be well approximated by the Pareto-distribution.[5] Provided that the minimum and mean burst sizes are known, the shape parameter α is given by

$$\alpha_{r/s} = \frac{\mu_{r/s,burst}}{\mu_{r/s,burst} - v^{min}_{r/s,burst}} \quad (2)$$

Typically, the minimum burst sizes can be set to the minimum allowed packet size, whereas the mean packet size can be obtained by usage measurements. In our model, the distribution of burst sizes is (unlike the burst arrival process) independent from the user activity pattern.

C. Simulation Setup

In a first step, we reproduce the conditions of the original dataset. Therefore, we consider a group of 310 individuals, and each of them is assigned a particular user type, such that all user types are represented to the same share as in the original dataset. We simulate the activity of one month, defined as 20 weekdays and 8 weekend-days.

Burst sizes have been set on the basis of [12]. Here, we consider only UDP and TCP data and define the minimum burst size as 52 Bytes (i.e. an empty UDP packet).[6] We simulate usage in 1 minute intervals over the entire period (session 1). Then we repeat the simulation at three different time scales $\Delta t[min] = \{1, 15, 60\}$, and reduce the total period to one day, to assess self-similarity (session 2).[7]

TABLE II
USAGE BURST PARAMETERS

Parameter	Received Bursts	Sent Bursts
Minimum Burst Size	116 Bytes	52 Bytes
Mean Burst Size	867 Bytes	387 Bytes

V. RESULTS

The simulation results (session 1) show that the model is able to reproduce usage characteristics on both, a user level and aggregate level. Figure 5 compares the simulated NAPs (solid lines) of the 3 largest user types (jointly representing 65% of the sample and $\geq$ 10% each) with the original cluster centroids (dotted lines).

The mean absolute percentage error (MAPE) for all patterns is 22% for received data and 16% for sent data, whereas the mean error (ME) is 0.0037 and 0.0038,

[5]Pareto PDF is $f_v(v) = \alpha \frac{v^{\alpha}_{min}}{v^{\alpha+1}}$ for $v \geq v_{min}$, and 0 otherwise.

[6]We chose a higher minimum value for received data, as 52 Bytes would lead to heavy distortions in the generation of Pareto-distributed random numbers.

[7]The model code is available at: https://github.com/swirsing/mts

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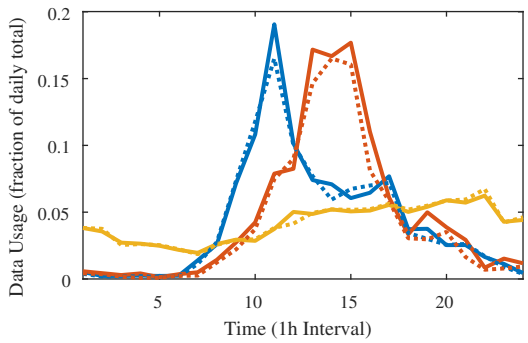

Fig. 5. Original (dotted) vs. Simulated NAPs (solid) for *weekday received* data usage of top user types (sample representation $\geq 10\%$)

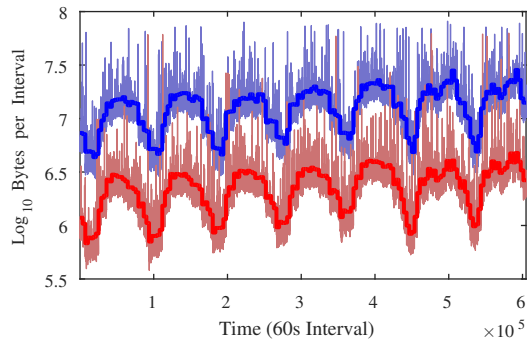

Fig. 7. Aggregate synthesized data usage ($\Delta t = 1min$, thin lines), hourly means (bold lines) for sent data (red) and received data (blue)

respectively. The corresponding average received volumes are depicted in Figure 6. While the larger MAPE indicates varying accuracy in the simulated values, the low ME suggests that these variations are widely balancing out over time. This observation is consistent with the empirical data, which also displayed high variances of the NAPs. It has to be noted that the ME is still 9% above the daily mean activity level, which is at 0.0417 ($= 1/24$). This is due to the fact that both, daily volumes and burst sizes are generated based on heavy-tailed distributions.

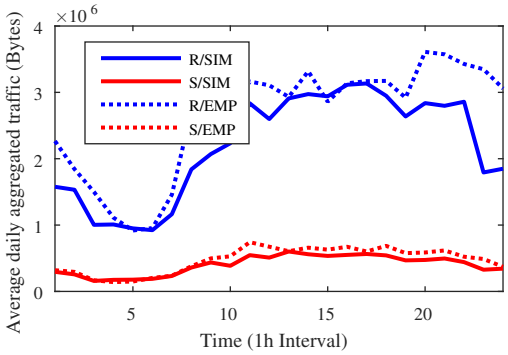

Fig. 6. Simulated (SIM) vs. empirical (EMP) received (R) and sent (S) data per average *weekday* and *user*

Figure 7 shows the aggregate usage sample of the first week simulated in session 1. The bold lines indicate the hourly mean volumes, which have been used to determine the λ-parameter of the non-homogeneous Poisson process, as defined in (1). The thin lines indicate the usage volumes within interval t_n. Although hourly means are preserved, the sample is strongly distorted (up to one order of magnitude) by the heavy-tailed burst size distribution. We did not apply volume or rate limits to obtain unbiased values. To model such constraints of real networks (rate limits, volume caps, network availability, congestion, etc.), post-processing algorithms must be applied on the obtained (raw) sequences.

Through the variation of the time scale Δt in session 2, we confirm self-similarity for all interval lengths. Specifically we calculate the Hurst parameters for each user sample at the different time scales. A Hurst parameter greater than 0.5 and approaching 1 indicate a high degree of self-similarity. Table III details the mean Hurst parameters over all users for the different interval lengths. We clearly find self-similarity in the range of several minutes which slightly decreases for hourly intervals.

TABLE III
HURST PARAMETER AT DIFFERENT TIME SCALES

Δt_n	1 min	15 min	60 min
Received	0.903	0.902	0.899
Sent	0.873	0.872	0.852

VI. CONCLUSIONS

In summary, we presented a granular and flexible mobile data usage model that we partly derived from empirical data. We illustrated that the model accurately reproduces the aggregate empirical patterns and allows arbitrary combinations of user types. Data usage volumes can be generated for simulation interval lengths ranging from one minute up to 1 hour. The model delivers raw received/sent volumes for the specified interval, without considering any network limitations. The latter can be achieved by post-processing algorithms of a network model. The combined use of such model with our data usage model will be helpful in evaluating the performance of future mobile communication systems. It can be used jointly with a subscriber mobility simulator, e.g. as presented in [13]. A limitation of the model is that it is not fully generative; it relies on empirical data such that the model should be periodically refitted on new empirical data to account for data usage growth over the long term.

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TABLE IV
MODEL DETAILS INCLUDING USER TYPE SPECIFICS

User Types (Weekday)							
User Type	Rec. Cluster	Sent Cluster	Num. Users	Rec. λ	Rec. β	Sent λ	Sent β
3-9	3	9	10	1.20E-07	4.93E-01	5.89E-07	5.47E-01
4-7	4	7	13	1.54E-08	5.20E-01	7.39E-08	7.07E-01
5-7	5	7	10	1.87E-08	7.16E-01	1.57E-07	7.21E-01
6-7	6	7	15	4.90E-08	8.91E-01	2.05E-07	9.95E-01
6-8	6	8	13	1.07E-07	6.20E-01	4.11E-07	7.40E-01
7-7	7	7	21	1.93E-08	9.20E-01	1.34E-07	9.77E-01
7-9	7	9	35	2.80E-08	6.57E-01	1.37E-07	6.46E-01
8-7	8	7	16	5.84E-08	1.25E+00	2.49E-07	1.07E+00
8-9	8	9	39	7.40E-08	6.37E-01	2.70E-07	6.77E-01
9-7	9	7	127	1.49E-08	6.41E-01	7.92E-08	6.95E-01
9-9	9	9	11	1.85E-08	9.34E-01	6.17E-08	7.15E-01
User Types (Weekend)							
User Type	Rec. Cluster	Sent Cluster	Num. Users	Rec. λ	Rec. β	Sent λ	Sent β
2-3	2	3	12	1.60E-07	0.47	6.52E-07	0.61
2-6	2	6	15	4.07E-08	0.77	2.07E-07	0.92
2-7	2	7	10	8.93E-08	0.78	4.03E-07	1.04
3-5	3	5	13	1.91E-08	0.55	7.96E-08	0.64
3-6	3	6	198	1.99E-08	0.58	9.86E-08	0.65
3-7	3	7	101	4.59E-08	0.60	1.68E-07	0.58

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